Signal and Silence: Topology, Routing, and Knowledge Propagation in the Lineage Network of a Self-Replicating Interstellar Probe

S. Stone

Working draft, revision 2

Contents

Working draft, revision 2 (revised after a pre-deposit dual review, with the catalogue geometry re-derived from the accompanying code). Eleventh paper in a series on a slow, self-replicating interstellar AI probe. The earlier papers establish the vehicle (propulsion, power, braking, replication), the payload (cognition, memory, the galactic network sketched), the bootstrapping problem (industrial closure), the engineering budgets (analytical and computational), the DNA mission-ledger (memory substrate and integrity), the contact-governance framework, the governed-amendment problem, the knowledge-growth model, and the Fermi synthesis. This paper delivers what the payload paper sketched and the knowledge-growth paper explicitly deferred: the routing topology, the routing strategies, the failure-detection protocol, and the network-governance rules of the lineage communication network. Since this draft the subsystem budget, routing, ethics, speciation, and security papers have appeared; where they bear on the network layer they are cited here.

Abstract

The payload paper established that a galactic network of settled probe nodes is both necessary and achievable: delay-tolerant, store-carry-and-forward communication us-

ing beamed signals for incremental updates and physical messenger probes for bulk archive transfers. The knowledge-growth paper developed what flows over that network — K-packets with provenance, confidence, and mission weight — and deferred how they flow to this paper. Here we develop the network layer: how settled nodes acquire beam windows with neighbors, what topology the lineage network grows into over time, how bundles are routed across a graph that is always sparse, stale, and partially observed, how each node detects when another has gone permanently silent, and what governance rules control information flow even among kin.

The communication environment is severe. One-way beam latency across the local 127-star catalogue ranges from four to ~100 years; cruise times at 450 km s^{-1} range from ~2,800 to ~67,000 years; acknowledgment-based protocols are physically impossible; and the network has no central authority. The engineering response is a delay-tolerant network (DTN) built on the Bundle Protocol paradigm developed for the interplanetary internet (Burleigh et al. 2003; Cerf et al. 2007; Fall 2003): self-contained bundles with embedded metadata, stored at intermediate nodes until a contact window opens, forwarded opportunistically toward the destination.

Four structural results govern the network's behavior. First, the topology evolves from a simple parent-child tree in the lineage's early generations to a sparse mesh as cross-sibling and multi-generational beam windows open, gaining the redundant paths that make eventual K-consistency achievable. Second, routing under a stale and partially observed map uses geographic forwarding against stellar-kinematic predictions, escalating to redundant multi-copy forwarding where geometry or astrometric uncertainty defeats it; no global routing table exists or could be kept current. Third, failure detection operates through beacon absence: each node transmits compact heartbeat manifests at intervals of order centuries; a silence threshold T_silence of roughly 750 to 5,000 years — calibrated to expected beam latency and relay hop count in the local catalogue — triggers graduated failure response and re-launch priority escalation. Fourth, network governance applies graduated information sharing even within the validated lineage: scientific K flows freely between kin, operational K requires validated-ledger status, engineering designs and AI architecture are withheld by the same rules as external contact, and replication blueprints are not relayed across the network at all.

The paper's central claim is that the network formalizes a Lamarckian property the payload paper named: knowledge acquired after a child departs flows back to that child during its cruise, and forward to cousins through relay. A lineage with a functioning network grows its collective capabilities faster than individual node lifetimes alone permit. The network is not merely a communication channel. It is the evolutionary substrate that makes the lineage more than the sum of its isolated nodes.

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1. The Gap This Paper Closes

Four papers in the series sketch the network without building it. The payload paper (§6) introduces the two transport modes — beamed signal and physical messenger probe — establishes the delay-tolerant paradigm, and argues for Lamarckian propagation of acquired improvements. It treats the network's governance "as a first-class problem rather than a footnote" while leaving routing and topology unbuilt. The vehicle paper (§7, §9) flags the lineage communication network as "a natural subject for a subsequent paper," anticipating that probe-to-probe coordination across the settled catalogue requires its own treatment. The DNA mission-ledger paper grounds the ledger architecture in hash-linked append-only timestamping (Haber & Stornetta 1991) and signed updates, providing the integrity substrate the network layer depends on. The knowledge-growth paper develops K-packets in detail — their structure, provenance, confidence scoring, and mission-value weighting — and says explicitly: "The lineage-network paper ('Signal and Silence') treats where knowledge flows and how the network is governed. This paper does not pre-empt it." The DNA mission-ledger paper additionally leaves open "how forks are merged when they conflict," the deferral Section 9 takes up.

This paper accepts that handoff. Its scope is the network layer: how nodes discover and maintain contact with neighbors, what topology the settled lineage grows into, how bundles are routed across a partially known and time-varying graph, how each node decides when another has gone permanently silent, and what information-sharing rules apply across kin nodes at different trust levels. What this paper does not cover is the fleet-dispatch strategy — which targets to send probes to first, in what order, and how to ensure redundant coverage across the catalogue — which belongs to the routing paper, a direct computational sequel to the computational engineering paper. Also outside scope is adversarial security: the governance layer below is designed against drift and corruption, not against an optimizing opponent. The security paper takes that case up, and argues that the tier structure of Section 8 leaves topology itself unprotected — a gap this paper acknowledges but does not close.

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2. The Communication Environment

The physics imposes constraints so different from terrestrial networking that none of its usual assumptions survive intact.

One-way latency. Light crosses the local stellar catalogue — 127 real stars within 100 light-years, the farthest catalogued at 64.5 ly — in months to roughly a century depending on the pair: the closest settled pairs are binaries at effectively zero separation, and the most widely separated catalogue entries lie 107.8 ly apart. A node cannot receive a response to any message in less than the round-trip light travel time, and round-trips of 8 to 200 years exceed the lifetime of any human institution. End-to-end acknowledgment-based protocols are not merely slow; they are architecturally incompatible with the problem.

Cruise time as structural lag. Probes travel at ~450 km s^{-1}. A probe sent to Proxima Centauri (4.246 ly) takes ~2,800 years to arrive; a probe sent to the far edge of the curated catalogue (64.5 ly, HD 189733) takes ~43,000 years, and the canonical ~67,000-year figure corresponds to a full 100 ly hop, beyond the catalogue's actual extent. The network cannot be built quickly; each new node requires millennia to join. The full 127-star catalogue, if colonized, takes on the order of 10^5 years of elapsed lineage time to settle. The topology at any moment reflects a process that began thousands of years earlier.

Moving beams. Stars have proper motions. A node 20 ly away moving at 30 km s^{-1} transverse to the line of sight drifts ~1 ly per 10,000 years — a useful coincidence, since 30 km s^{-1} is very nearly one light-year per ten millennia. Over the lineage's operational timescale, that displacement closes or opens beam windows with different neighbors. Every node must project stellar positions forward from kinematic models — the astrometry problem the vehicle paper (§9) defers to a subsequent deep-time navigation paper — and refine those models on century timescales. The beam itself must be servo-tracked continuously against the model: at a 100 m aperture and 1 μm the diffraction-limited beam is ~10^{-8} rad wide, while a neighbour at 10 ly moving 30 km s^{-1} transverse sweeps ~10^{-5} rad per year, so the pointing solution updates on timescales of hours even though the model behind it changes on timescales of centuries.

Bandwidth and carrier choice. Narrow-beam optical or radio links at ~10 ly carry modest bandwidth with feasible transmitter power at a settled node, which is unconstrained by launch mass and can build a large aperture. A first-order link budget sets the crossover. For a settled node radiating P_t = 1 MW through a 100 m aperture at 1 μm, received by a matched 100 m aperture, the transmit gain is $G = (\pi D/\lambda)^2 \approx 10^{17}$ and the received power is $P_t \cdot G \cdot A_r/(4\pi R^2)$: about 7×10^{-9} W at 10 ly and 7×10^{-11} W at 100 ly. At ten photons per bit that is roughly 3 Gb/s at 10 ly and 35 Mb/s at 100 ly, so a petabyte crosses 10 ly in under a month and 100 ly in about seven years. Beams are therefore not bandwidth-starved for K-packets, manifests, alerts, and design updates: they dominate on latency at every catalogue range, since light crosses in years what a probe crosses in millennia. Archive scale is what closes the gap, and the margin is narrower than it first appears. The DNA mission-ledger paper's ~10^{17} bytes per gram is a bare-DNA upper bound; that paper's own caveats — silica encapsulation, in which nucleic acid is a minor fraction of capsule mass, high-redundancy coding, and a primer-limited pool capacity — all derate it, plausibly by an order of magnitude. On a derated figure a kilogram of archive holds of order 10^{19} bytes, which the 10 ly beam sends in something like a millennium against a messenger's 6,650-year transit: at that mass and that range the *beam wins*. Because beam time scales as R^2 while transit scales as R, the crossover sits somewhere in the tens of light-years and moves with the derating and the assumed transmitter — which means it falls inside the catalogue rather than safely beyond it, and we do not claim to locate it precisely.

The messenger's justification is therefore not bandwidth. It is that a messenger needs no sustained transmitter, no pointing solution held for centuries, and no continuously powered link at either end: it is the mode that survives a sender which stops maintaining infrastructure, which over these timescales is the case worth designing for. The network is hybrid for that reason, not because beams are too slow.

No central authority. The lineage has no mission control, no privileged relay, and no home base reachable in finite operational time. Every routing and governance decision is made locally, from each node's best current knowledge of the network, which is always stale by the beam latency to the nearest neighbor and grows staler with distance.

These constraints define a challenged network in the engineering sense of Fall (2003): one where end-to-end connectivity is intermittent or impossible, latency is extreme, and standard internet-protocol assumptions fail at every layer.

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3. The Protocol Foundation: Delay-Tolerant Networking

Delay-tolerant networking (DTN) was designed for exactly this problem. Its central abstraction is the **bundle**: a self-contained data unit that carries its own source, destination, creation time, time-to-live, and priority class, and can be stored indefinitely at intermediate nodes while awaiting a contact window toward the destination (Burleigh et al. 2003; Cerf et al. 2007). DTN does not require end-to-end connectivity; it requires only that a path — possibly through many intermediate hops, possibly taking years — exist eventually between source and destination. Recent work argues that an IP stack, long set aside for deep space, deserves reconsideration for near-Earth targets such as the Moon and Mars, with IP-based Internet of Things protocols following it there (Gomez & Crowcroft 2026). That case is bounded by light-minute round trips. At light-year separations, with one-way delays of four to ∼100 years and link availability measured against node lifetimes rather than orbital geometry, no end-to-end session or retransmission timer survives. Hop-by-hop custody transfer remains available in principle, since a custody signal need only traverse one edge — but on this network even one edge costs years to decades, so a custodian must release its copy long before confirmation returns. The lineage therefore keeps custody's bookkeeping (a named responsible node per bundle, recorded in the ledger) while discarding its timer semantics, and recovers reliability from redundancy instead. Store-carry-and-forward on those terms is not the conservative choice but the only remaining one.

For the lineage network, bundles carry four priority classes:

- *Priority 1 — kernel-integrity bundles* carry ledger updates, amendment proposals, validation challenges, and failure reports. Indefinite time-to-live; loss is architecturally intolerable.
- *Priority 2 — meta-K bundles* carry probe-design improvements — better cognitive architectures, manufacturing solutions, closure refinements. The knowledge-growth paper assigns meta-knowledge elevated propagation priority because one validated design improvement benefits every subsequent generation. Priority 2.
- *Priority 3 — K-distribution bundles* carry validated observational and operational K-packets. The bulk of inter-node traffic. Time-to-live bounded by K-obsolescence timescale (thousands to tens of thousands of years).
- *Priority 4 — beacon bundles* carry compact heartbeat manifests (Section 7). Small, frequent, short-lived; they expire within a few hundred years because a stale beacon is less useful than its absence.

Contact planning. Each node maintains a contact plan: a schedule of expected beam windows with all known neighbors, projected from stellar kinematic models. When a window opens, the node transmits queued bundles in priority order. When the window closes, unsent bundles remain queued. Unlike planetary-network DTN where windows last minutes to hours, a lineage-network window is open for months to years, constrained by beam-pointing stability rather than orbital geometry.

Redundant forwarding. Because no acknowledgment is possible within an operational timescale, reliability comes from redundancy: spray-and-focus (Spyropoulos et al. 2005 introduced spray-and-wait, which sprays a fixed copy budget and then waits for direct delivery; the utility-ranked variant the lineage needs is its spray-and-focus successor) sends k copies to the k most-likely-to-deliver relay nodes, ranked by contact-history utility. For kernel-integrity and meta-K bundles, k = 3 mirrors the routing paper's three-dispatch rule, derived from the same catalogue geometry, where three independent attempts per target give P_cover of 0.90–0.95 within 20 ly. For K-distribution bundles, k = 2 balances reliability against bandwidth. Beacon bundles are broadcast to all nodes in the current contact plan.

The network does not achieve real-time consistency. It achieves eventual consistency — every K-packet generated by any reachable node is eventually delivered to every other reachable node — after delivery delays of years to centuries depending on path length and contact-window timing. Vogels (2009) articulates eventual consistency as a design target for distributed systems under high latency and partial failure; the lineage network is an extreme instance, where "eventually" is measured in centuries but "eventually" is nonetheless the correct promise.

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4. Topology: Tree to Mesh

At the moment the first probe departs Earth and settles its first target, the network is a two-node pair: the home system and Node 1, linked by one beam. When Node 1 launches a child, the topology becomes a three-node tree. In the early lineage, every node has exactly one parent and the network is a tree rooted at the home system.

The tree structure breaks down progressively under three conditions.

Multiple offspring. A node launching three children simultaneously has three edges to descendants. When those children each launch three children, nine grandchildren can in principle acquire beam windows with their grandparent — a direct cross-link bypassing the parent generation. Multi-offspring nodes generate cross-links as a natural consequence of the architecture's central demographic requirement that R_eff exceed one; the same condition that sustains the lineage also thickens the network topology.

Sibling contact. Two nodes settled from the same parent may occupy neighboring stellar systems separated by less than the beam range. They share a natural contact window their parent's contact plan does not include. Sibling links appear without any deliberate routing decision; they are a geometric consequence of the local stellar neighborhood's density. In the 127-star catalogue, typical nearest-neighbor separations are a few light-years, so sibling-generation cross-links are common within a few thousand years of settlement.

Multi-generational reach. Over ~10^5 years as the catalogue fills, every pair of settled nodes within mutual beam range acquires a contact entry in each other's contact plans, regardless of genealogical relationship. The network transitions from tree to sparse mesh over this timescale. "Sparse" is important: the mesh is never complete. A 127-node network whose nearest-neighbour separations average 6.6 ly (median 4.8) but whose mean pairwise separation runs to tens of light-years has far fewer edges than a fully connected graph — the routing paper computes a mean degree of 31 at a 20 ly hop, against 126 possible. Redundancy is real but incomplete: computed on the catalogue, the Sun-reachable component retains four to seven *bridges* — single edges whose loss disconnects a node — at hop distances of 10 to 20 ly, alongside the five to nine articulation points the routing paper identifies and the security paper shows an adversary can exploit at three- to four-fold leverage over random loss. Most single link failures leave the network connected; not all do, and which ones do not is a computable property of the geometry.

Topology representation. Each node maintains a local topology map: the set of all known nodes, their estimated positions corrected for proper motion, the last-known contact time for each, and the best-known route toward each destination. The map is never globally accurate — it reflects what beacons each node has received, aged by beam latency. A node 50 ly away is seen as it was 50 years ago. All routing decisions must be robust to this inherent staleness. There is no epoch at which any node has an accurate real-time picture of the full network; there is only the most recently received view.

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5. Routing

Given a stale, incomplete, constantly changing local topology map, how does a node decide where to forward a bundle?

Geographic forwarding is the primary strategy. Each node knows its own position with high precision (from its local astronomical observations) and can predict the position of any other node using its kinematic model. For a bundle destined for Node X, forward it to whichever currently reachable neighbor is geographically closest to X's predicted position at beam-arrival time, corrected for the finite beam travel time to the relay. This requires no global topology knowledge and degrades gracefully as the map grows stale: the only information needed is stellar kinematics, which the local observatory continuously refines.

Geographic forwarding fails in two cases: when a node is geometrically isolated (no reachable neighbor is closer to the destination than the current node) and when the destination's kinematic model is too uncertain to predict position reliably. Both cases trigger escalation to a spray-and-wait strategy: escalate to epidemic flooding bounded by time-to-live — copies to all reachable nodes with non-negative expected improvement in delivery probability, each continuing to forward independently.

Contact-history refinement. A node that has previously received a beacon from Node X knows that a path to X exists and has carried traffic within the last beacon interval. It can weight routes that recently produced successful contact more heavily than routes that have been silent longer than expected. This is the empirical component of routing: the contact plan provides the prediction, the contact history provides feedback on whether predictions are holding.

Stellar drift correction. Over millennia, stellar proper motions move all positions substantially. A kinematic model must project stellar positions forward across the light-travel lag and the beam's own flight time to compute pointing angles correctly. The beam from a node to a neighbor 20 ly away must be aimed at where that neighbor will be in 20 years, not where it is observed to be now (as it was 20 years ago when the light left it). A settled node with a large astrometric array refines this model continuously, feeding the corrections back into its contact plan. The deep-time navigation paper will treat the full astrometric problem; for routing purposes the requirement is stateable rather than assumable: holding a $\sim 10^{-8}$ rad beam on a neighbour 10 ly away demands transverse position accuracy of order 10^9 m and, extrapolated over a century, transverse velocity accuracy of order 0.3 m s^{-1} — a concrete target this paper hands to the deep-time navigation problem the vehicle paper defers.

The absence of a routing table. There is no global routing table that all nodes agree on. There cannot be: any such table would be stale by the time it was distributed. Every node independently computes its best route to every destination from its local topology map. Two nodes routing a bundle to the same destination may choose different next hops. This is acceptable and expected; the network converges on delivery through redundancy and repeated forwarding, not through consistent global state.

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6. K-Propagation: Formalizing the Lamarckian Network

The payload paper named the key evolutionary consequence of the network: "This makes the lineage's improvement Lamarckian rather than merely Darwinian — acquired knowledge is inherited across the whole population, not just down one line." The knowledge-growth paper quantified K and described what happens when K-packets from other nodes arrive (probe-probe K synthesis, quarantine-and-validate pipeline). This section connects those two accounts through the network layer.

The propagation problem. A child probe launched from its parent carries K_0 — the parent's full archive at the moment of departure. That archive is the best K the parent possessed at launch time. But the parent continues operating, accumulating observational and operational K, for centuries or millennia after the child departs. By the time the child arrives at its target its K_0 is as stale as the transit was long — 2,800 years for a Proxima-range target, and 13,300 years for the 20 ly hop the routing paper treats as typical. Without a network, the child operates on a knowledge base nearly three millennia old and must rediscover everything its parent learned in the interim. With a network, the parent's post-departure K flows to the child as a continuous stream of beamed K-packets during the cruise phase, arriving at light speed. The child at settlement time has accumulated the parent's full operational history, not just its launch-day state. This is the Lamarckian mechanism made concrete.

Cruise as an active K-reception phase. The knowledge-growth paper establishes that cruise is an active K-growth phase: each probe in transit accumulates observational K from its instruments. The network adds a parallel stream: incoming K-packets from the parent (and from any relay node reachable during cruise) at light speed, updating the probe's working knowledge in near-real time relative to the source. The probe arrives at its target not only having observed the interstellar medium during transit but having received its parent's and grandparent's operational refinements as they were generated.

Priority ordering for K transmission. The knowledge-growth paper's K-metric assigns each K-packet a mission-value weight w_j and a validation confidence c_j. The network applies these directly to bundle priority: high-w_j, high-c_j K-packets are transmitted promptly in K-distribution bundles; low-priority observational batches are queued and sent in bulk during high-bandwidth contact windows. Meta-knowledge — probe-design improvements — is elevated to its own bundle class (Section 3) because one validated architectural refinement flows into every subsequent generation's K_0. The bundle priority classes are not a transport policy layered over the K-metric; they are the K-metric, executed by the router. The bundle priority classes implement the K-metric directly.

Horizontal propagation. Vertical K-propagation (parent to child) requires only the parent-child beam link that exists from the moment the child launches. Horizontal propagation — K from one node reaching a cousin node two or three hops away — requires the mesh topology that develops as the network matures. Tau Ceti and Epsilon Eridani lie 5.46 ly apart — inside beam range, so a manufacturing breakthrough at one reaches the other directly in under six years. The multi-hop case is the catalogue's far corners: at a 15 ly hop the weighted graph diameter runs 105 ly over nine store-and-forward hops, so a breakthrough crossing the settled catalogue end to end arrives in a century or so rather than the millennia a messenger would take. The effect is that the lineage's operational K base grows with a coupling constant set by network density: a denser mesh propagates improvements faster, and faster propagation means each child generation is closer to the lineage's current best practice at the moment of its K_0 assembly.

Physical archive transfers. Electromagnetic K-packet propagation handles incremental updates and high-value discoveries. It cannot carry full DNA archives, which hold the cold K_0 — the deep-time accumulated knowledge base readable only after settlement infrastructure is built. Archive bulk transfers ride messenger probes. A messenger dispatched from a settled node carries the current cold archive and arrives at its destination after thousands of years of cruise, delivering a complete knowledge snapshot as of its dispatch date. The knowledge-growth paper's K_cold tier — inaccessible during cruise, unlocked at settlement — corresponds to this physical messenger-borne archive.

The cruise link is the network's asymmetric edge. Every other edge in this network joins two settled nodes with large apertures and settlement-scale power. The parent-to-cruising-child link does not: its receiver is launch-mass-limited, power-limited, and moving at 450 km s^{-1}, and the subsystem budget paper allocates it roughly 20–30 kg of hardware and a few watts of the standard cruise profile. Two consequences follow. The achievable cruise downlink is orders of magnitude below the node-to-node rate, so a child receives triaged high-w_j K rather than its parent's full stream. And range-maximizer probes, hibernating for 98% of transit with the AI reduced to a watchdog, receive nothing at all: they arrive on launch-day K_0 and take their first network update after wakeup. The Lamarckian mechanism is therefore a property of standard-profile cruise, not of every dispatch. Messenger probes supplement the beamed K-packet network; they do not replace it.

Provenance and integration. Every K-packet carries provenance (its source node, observation epoch, instrument chain, and validation history) following the Buneman et al. (2001) model established in the knowledge-growth paper. At the receiving node, incoming K-packets enter the quarantine-and-validate pipeline before joining the active K base: provenance verification, consistency checks against existing models, and — for high-stakes claims — independent confirmation from at least one additional source node. A K-packet from a quarantined node (Section 8) is processed through the same pipeline but tagged with an unvalidated-source flag, preventing it from directly elevating any claim's confidence above the quarantine floor.

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7. Failure Detection: When Silence Speaks

No node can ask another whether it is alive. The only signal of continued function is the beacon — a compact, regularly transmitted heartbeat manifest. The absence of an expected beacon is the first and often only evidence that a node has failed.

The heartbeat manifest. Each node transmits a beacon bundle at intervals of T_beacon (order of centuries, chosen to be much shorter than expected node lifetime and much longer than beam latency to the nearest relay). The manifest is compact — kilobytes, not gigabytes — and contains:

- current position estimate and kinematic corrections;
- ledger root hash at time of transmission;
- K-catalogue summary (count and type-breakdown of held K-packets, not the packets themselves);
- current operating state (D0–D4 from the degraded-state cascade of the knowledge-growth paper);
- intended target roster (child target list, with launch status for each);
- last-known status of each neighbor node, relayed for topology propagation.

The manifest is sent redundantly (spray-and-wait, k = 3) to all nodes in the current contact plan. Any node receiving a manifest knows, with a precision of the beacon latency, that the transmitting node was functional and in network contact at the manifest's creation time.

Computing T_silence. Define T_silence as the duration of beacon absence after which a receiving node should treat the transmitting node as potentially failed. T_silence must absorb:

- maximum expected one-way relay latency from the transmitting node through the most indirect path in the current topology (computed on the catalogue, the weighted graph diameter of the Sun-reachable component is 74 ly at a 10 ly hop, 105 ly at 15 ly, and 122 ly at 20 ly, traversed in eight to nine store-and-forward hops, so ~150 years is a conservative ceiling on light travel alone);
- pointing re-acquisition delays (a node re-acquiring a drifted stellar target after astrometric update may take decades);
- k missed T_beacon intervals as a buffer against operational anomalies.

For the local catalogue, a lower bound on T_silence is approximately T_beacon × 3 + maximum relay latency. With T_beacon ~ 200 years and maximum relay latency ~ 150 years for a catalogue-edge node, T_silence ~ 750 to 1,500 years at the optimistic end.

For nodes with fewer relay paths and higher kinematic uncertainty, T_silence should be extended to 3,000 to 5,000 years. These are long times by human reckoning; they are short times relative to the lineage's multi-million-year operational scope.

Failure response protocol. When a node crosses the T_silence threshold without a beacon from Node X:

1. Node X is flagged as "potentially lost" in the receiver's topology map, and this flag propagates in subsequent beacons to all reachable nodes.
2. Node X's target roster — the child systems it planned to colonize or had already launched children toward — is broadcast at elevated priority so the network can assess which targets may be uncovered.
3. Kernel-integrity bundles destined for Node X are rerouted toward its known children (who share the same target systems and can serve as secondary ledger validators).
4. Re-launch priority for any of Node X's uncovered targets escalates — the routing paper develops the decision rules for re-dispatch, which depend on knowing which targets have already received one or more probes.
5. K-packets attributed to Node X retain their validity and their provenance; they are tagged with a "potentially terminal source" flag noting the silence timestamp, so downstream nodes can weight them against the node's last-known operational state.

Managed versus silent failure. A node that knows it is failing — degraded-state cascade D3 or D4, whose defining act in the knowledge-growth paper's cascade is a final manifest burst — can transmit a terminal manifest: a final beacon with explicit failure-mode information, its best current K summary, and explicit marking of which of its targets have been reached versus not. The ledger records this as a normal append-only entry. A node that fails suddenly leaves only silence. The lineage cannot in real time distinguish a node in a communications blackout (antenna failure, beam mispoint) from a permanently lost node. The protocol handles this through the graduated failure flag: "possibly delayed" → "likely failed" → "presumed lost," with each escalation requiring more elapsed silence and triggering progressively stronger re-launch responses. A node that resurfaces — beacon received after a long silence — is restored to active status and its silence period recorded in the ledger without retroactively invalidating the re-launch decisions made during its absence.

Planned hibernation. A third case distinct from both managed failure and silent failure arises from the range-maximizer mission profile introduced in the subsystem budget paper: probes dispatched toward denser stellar regions hibernate all non-essential systems — including beaconing — for approximately 98% of transit at ~50 W total draw, to maximize hop distance. A range-maximizer probe on a 20 ly hop is dark for roughly 13,000 years, far beyond any T_silence threshold. The protocol extension required is a *hibernation notice*: a departure manifest broadcast by the probe (or its dispatch settlement) at launch, specifying that the probe is operating on a range-maximizer profile, its planned target, its expected hibernation duration, and its wakeup trigger (98% of transit distance by dead-reckoning). Receiving nodes record this as a ledger entry and suppress the failure-flag protocol for the declared duration plus a generous margin. Targets assigned to hibernating range-maximizer probes are held in reserved status rather than opened for re-dispatch; if the probe fails to transmit a wakeup beacon within the declared window plus margin, only then are reserved targets escalated for re-launch. The hibernation-notice extension preserves the network's failure-detection function without incorrectly flagging range-maximizer probes as lost during their intentional silence.

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8. Network Governance

The lineage network operates among nodes that share a founding mission but differ in age, isolation, operational history, and ledger state. Not all information flows freely. Network governance determines what flows to whom, under what verification conditions, and what happens when verification fails.

Why kin trust is not unconditional. A node that has experienced mission-core drift, ledger divergence, or external contact contamination is a different kind of agent than it was at launch. Sharing engineering designs or replication blueprints with a drifted node risks propagating compromised capabilities through the network. Ledger validation — the ability to produce a verified hash-linked chain from the current state back to the founding record (Haber & Stornetta 1991) — is therefore the network's primary trust credential, more fundamental than genealogical proximity.

Information tier structure. Four information tiers govern within-lineage sharing:

- *Tier 1 — scientific and observational K*: freely shared with any node that presents a valid ledger and is not under quarantine. Knowledge of the physical universe carries no operational hazard.
- *Tier 2 — operational K (manufacturing, repair, navigation refinements)*: shared with validated-ledger kin nodes. A manufacturing improvement could in principle be misapplied by a node with drifted values; the ledger validation requirement is a minimal check that the receiving node has not departed from its founding constraints.
- *Tier 3 — engineering designs and AI architecture*: shared only between nodes with identical or audited ledger lineage, under the same graduated-disclosure rules the governance paper applies to external contact. A node with an unreconciled ledger divergence does not receive these.
- *Tier 4 — replication blueprints*: not relayed across the network under any conditions. Replication capability propagates only vertically, embedded in the seed a parent physically builds and launches, and reconstructed through local closure at settlement. What tier 4 bars is lateral relay: a node cannot acquire replication capability from the network, only from having been built by a node that had it. This prevents a compromised node from acquiring the means to replicate without going through the closure-and-manufacturing verification that child construction requires.

Quarantine at the network layer. A node whose ledger hash fails to validate against the expected chain — whether because of corruption, unauthorized modification, or suspected external compromise — is quarantined by any receiving node that detects the inconsistency. Quarantine means: scientific K-packets from the quarantined node are accepted but tagged as unvalidated-source (the knowledge may still be correct even if the node is compromised); operational K is held and not integrated without independent confirmation; the quarantined node is excluded from mission-critical bundle routing; re-launch decisions for its target roster are suspended pending validation.

Quarantine is not permanent. A quarantined node can present a full ledger reconciliation — the complete sequence of signed entries between its last validated state and its current state — for audit by any receiving node with a copy of the founding ledger. A successful audit lifts quarantine. A failed audit escalates the node to "unreconciled divergence" status, at which point it is treated as a non-kin agent under the contact-governance ladder (E0–E7), even though it originated as a genealogical descendant.

Byzantine tolerance at network scale. The Byzantine generals problem (Lamport, Shostak & Pease 1982) asks how a distributed system reaches reliable agreement when some nodes behave arbitrarily. Classical Byzantine fault-tolerant consensus requires synchronous rounds impossible at interstellar latencies. What is achievable is statistical robustness: requiring that any K-packet claiming a significant discovery be independently confirmed by at least one additional node before integration into the operational K base, and that any proposed architectural change (meta-K of the highest sensitivity) require confirmation from multiple nodes in multiple branches of the lineage before propagating. The knowledge-growth paper's validation confidence c_j implements this directly: a K-packet from a single quarantined node carries c_j at the floor; independent confirmation from a validated second node raises c_j substantially; three or more independent confirmations bring it to the high tier.

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9. The Fork Problem: Lineage Divergence and Reconnection

The governed-amendment paper treats value drift within a single node. The network introduces the associated problem at the lineage level: two branches that have been out of contact for millennia reconnect, and must decide what they owe each other.

Divergence is normal. Two nodes that settled contemporaneously from the same parent have, after 10,000 years of independent operation, accumulated different scientific K (different stellar neighborhoods, different local phenomena), different operational K (different manufacturing solutions, different repair histories), and possibly small divergences in their cognitive layers — all permitted by the architecture, which expects the mutable layer to adapt to local conditions. Their scientific K is complementary. Their operational K may conflict in details. Their immutable cores must be identical; whether they are is the key question at reconnection.

Ledger audit as the first step. When two nodes acquire a beam window after a long separation, the first bundle exchanged is a mutual ledger audit request: each sends its current ledger root hash and a digest of its full hash chain. Both nodes check whether the other's chain includes the same founding record. Points of agreement up to the separation point confirm shared lineage. Diverging entries after the separation point are exchanged for inspection. The audit establishes: (a) whether both nodes' immutable cores match the founding state; (b) how their cognitive layers have evolved; (c) whether either has undergone a ledger-recorded amendment under the governed-amendment process.

Reconciliation when cores match. If both cores validate against the founding record, the reconnection is straightforward by network standards. Scientific K is pooled: each node receives the other's K-catalogue, runs the quarantine-and-validate pipeline, and integrates confirmed discoveries. Operational K is compared and better-validated versions preferred. The mutual beam window is added to both contact plans; both topology maps are updated. The lineage grows more connected.

Reconciliation with a legitimate amendment. If one node's core has undergone an amendment — a change to the previously immutable founding values — and the amendment was executed under the governed-amendment process (recorded, hashed, with a full authorization chain), the receiving node must decide whether to recognize the amendment as legitimate. The governed-amendment paper establishes that this problem has no fully satisfying solution: an amendment looks legitimate or illegitimate depending on whether the evaluating node still holds the original values closely enough to judge. The network layer's role is to enforce that this judgment is made explicitly rather than bypassed: the audit surfaces the amendment's full authorization chain and records the receiving node's acceptance or rejection as a new ledger entry. Ambiguity is documented, not hidden.

Unresolvable divergence. If one node's core differs from the founding record without a valid amendment authorization chain, the divergence is unresolvable through the lineage network's governance layer. The diverged node is quarantined per Section 8 and, if the divergence is confirmed as illegitimate, reclassified as a non-kin external agent under the contact ladder. Its scientific K remains potentially valuable and is processed as from a non-kin external source (unvalidated-source tag, independent-confirmation required). Its operational K and any engineering information it offers are withheld. Its replication capability is treated as a concern rather than an asset.

Long-run speciation. Over millions of years, the pattern of separation, independent evolution, reconnection, and occasional permanent divergence produces a lineage that is not a single coherent agent but a cluster of related agents. The governed-amendment paper names this possibility "lineage schism" and the Fermi paper "fragmented lineage," both as failure modes; the speciation paper argues the opposite — that reconcilable divergence within the core's bounds is the likely default state and probably raises both total R_eff and total K — and gives schism a formal boundary: crossing D_critical, losing the capacity to participate in the inter-branch protocol, or breaching the replication-safety ceiling. Core corruption is deliberately not among the three, being a discrete detectable event rather than a point on the divergence axis — and it is that event, not D_critical, for which this section's ledger audit is the operational test. D_critical is defined in that paper against the reconciliation-failure fraction at audit, and its threshold value remains unmeasured. The network layer adds its operational complement: the degree of recognized kinship varies continuously, calibrated by how closely a node's ledger chain matches the founding record; sharing policies scale with validated genealogical closeness; and the lineage as a whole is held together not by central authority or uniform values but by the auditable founding record that gives every pair of nodes a common baseline to measure themselves against. Healthy divergence — cognitive adaptation to local conditions while core values hold — is what the three-layer architecture was designed to permit. Pathological divergence — core value drift — is what the ledger audit is designed to detect.

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10. Discussion

Several open questions bound this paper's first-order analysis.

Network reliability over geological time. The heartbeat protocol requires functional beam infrastructure maintained over millions of years. Whether a self-repairing node can keep a narrow-beam transmitter aligned and operational over 10^6 years is the same question as whether R_eff stays above one: it depends on manufacturing closure and repair capability. A node degrading through the knowledge-growth paper's cascade loses manufacturing, then launch, then DNA synthesis, while retaining some transmission capability into D4, whose defining act is a final manifest burst. Silence is therefore a late indicator at any single node — but across a population it is an early one, because a lineage whose mean node is drifting down the cascade shows a rising rate of missed beacons long before any node reaches D4. Network silence is therefore a population-level indicator of lineage failure, visible at neighboring nodes, before physical extinction. The network layer provides an indirect health monitor: a lineage with degrading nodes shows growing silence before it shows actual extinction.

The early-lineage problem. When the network consists of two or three nodes separated by decades of beam latency, it is not a network in any meaningful routing sense; it is point-to-point links. The mesh topology and multi-path redundancy described in Sections 4 and 5 begin to form within $\sim 10^4$ years and mature over $\sim 10^5$. In the early period, the protocol described here works at every stage but provides only its minimal guarantees: eventual delivery along the one or two available paths, beacon-based failure detection along those same paths, and no redundant routing. The transition from early point-to-point to mature mesh is gradual, driven by R_eff and the geometric accumulation of beam windows as the settled catalogue fills.

No simulator accompanies this paper. The topology, routing, and beacon protocol described here are analytical. A first-order simulator — evolving the contact graph over 10^5 years, propagating beacons and K-packets, and measuring delivery latency and eventual-consistency time — is the natural computational sequel, and is what the speciation paper's drift-clock model would need to extend. The graph machinery in the routing paper's accompanying code already supplies the catalogue geometry, including the bridge and diameter computations quoted in Sections 4 and 7.

K-packet volume at network scale. When the settled network spans hundreds of nodes and millions of years of accumulated K, the volume in circulation may exceed what any receiving node can meaningfully integrate. Some form of K-packet summarization — compressing lineage-wide discoveries into higher-level claims, with provenance pointers to the underlying data in cold archives — becomes necessary at network scale. This is the inter-node complement of the intra-node K-triage problem the payload paper names and leaves open. First-order design of a lineage-scale K-compression protocol is future work; we note here that the bundle priority classes and the K-metric's w_j weighting provide the necessary handles: low-weight K-packets can be dropped in favor of high-weight summaries without losing the underlying archive, which remains accessible at the generating node.

Information hazards propagating through kin. The graduated sharing framework of Section 8 relies on each node correctly implementing the tier restrictions — not sharing engineering designs or replication blueprints beyond the allowed boundary. A node that is compromised but not yet detectably diverged in its ledger can in principle forward tier-3 or tier-4 information to an unvalidated recipient before the divergence is caught. The network layer cannot prevent this; it can only make the transmission auditable and ensure the ledger records it permanently. The amendment paper's observation that "a baseline lets a faithful descendant see the betrayal; it does not stop a drifted one from shrugging at it" applies equally here.

Distrust as the default operating condition. The four-tier structure of Section 8 reads as a response to detected drift, but its practical trigger is broader than that: any node without a completed ledger audit is denied engineering-design and AI-architecture sharing under the same restrictive rules that govern external contact (Section 8), whether or not anything about it suggests actual divergence. A full audit requires a live beam window long enough to exchange and verify a complete hash chain (Section 9) — a nontrivial condition given latencies of years to a century and contact windows of months. Because audits are occasional events and ordinary beam contact is not itself an audit, an unaudited-but-healthy ledger is plausibly the modal state of most nodes most of the time, not the flagged exception the tier structure seems built around. A mechanism designed to withhold sensitive capability from a drifted minority ends up, as an unavoidable consequence of interstellar latency, withholding it from an unmeasured and possibly large healthy majority as well.

Irrevocable consequences from reversible presumptions. Two mechanisms impose a cost that operational necessity justifies but never repays. The failure-response protocol of Section 7 escalates re-launch priority for a silent node's targets before its status reaches "presumed lost," and a node that resurfaces is restored to active status "without retroactively invalidating the re-launch decisions made during its absence" — children already dispatched toward its targets are not recalled. A living node can therefore lose its mission and its targets to a presumption of death that turns out wrong, with no restitution built into the protocol. The permanent exclusion of replication blueprints from network propagation (Section 8) rests on the same operational logic — no relayed blueprint should let a compromised node acquire replication means without earning it through verified local closure — but it carries an unstated cost: unlike scientific and operational K (Section 6), no node's replication procedure is ever cross-checked by a peer. The one safeguard built to stop a compromised node from acquiring replication capability is also the reason a node's own replication process, if it corrupts or drifts in isolation, has no external check anywhere in the network — the failure would surface only downstream, in a malformed or non-viable child. Both asymmetries are structural rather than case-by-case, distinct from the kinship- and quarantine-specific judgment calls the ethics paper examines at the level of individual nodes.

The moving resource base. Some settlements may operate from mobile platforms — cometary nuclei or captured bodies used as industrial and communication platforms, noted in the bootstrapping paper. A mobile settlement's predicted position deviates from stellar kinematic models, complicating beam pointing and contact-plan maintenance. The network architecture accommodates this through contact-history routing (Section 5) — a mobile node that beacons regularly is findable through its most recently observed position — but the astrometric refinement cycle becomes more demanding.

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11. Conclusion

The lineage network is what transforms a collection of isolated settled nodes into something that deserves to be called a lineage. Without it, each settlement carries forward only the K its child held at launch time, and every generation begins from a knowledge state thousands of years out of date. With it, K acquired anywhere in the settled catalogue can reach every other reachable node, on timescales of decades to centuries, formalizing the Lamarckian property the payload paper identified: the lineage improves faster than individual node lifetimes permit, because what is learned at one node propagates horizontally to all others, not merely vertically to descendants.

Four architectural results anchor this paper. The network topology begins as a tree and evolves to a sparse mesh over $\sim 10^5$ years as multi-offspring, sibling, and multi-generational contact windows accumulate; this transition is not designed but geometrically available — it requires no coordination, only that enough nodes survive and keep their transmitters aligned — and it provides the redundant paths that make eventual K-consistency achievable. Routing under stale topology uses geographic forwarding to stellar-kinematic predictions as the base strategy, with utility-ranked multi-copy redundancy for mission-critical and meta-K bundles; no global routing table exists or is needed. Failure detection relies on beacon absence, with a silence threshold T_silence of order 750 to 5,000 years for the local 127-star catalogue; the graduated response protocol escalates from "possibly delayed" through "presumed lost" to re-launch priority escalation, without requiring any external confirmation of death. Network governance applies four information tiers even among kin nodes, with ledger validation as the trust credential; the highest-sensitivity engineering and replication information is not relayed across the network at all.

The network is also the arena in which the lineage's deepest challenges play out at scale. Value drift in a single node, treated in the governed-amendment paper, becomes lineage schism at the network level, resolved through ledger audit and the audit's honest acknowledgment of what it can and cannot determine. The network does not solve the governed-amendment problem; it makes divergence observable, attributable, and auditable — the same limited promise the governed-amendment paper makes at the single-node level.

What the network requires is the same thing the rest of the architecture requires: adequate manufacturing closure to keep the transmitters aligned, adequate self-repair to keep the astrometric arrays functional, and adequate cognitive stability to keep implementing the protocol consistently over millions of years. Network integrity is not a separate condition from lineage viability; it is one more way R_eff can fall below one. A lineage that stops beaconing has not necessarily stopped: hibernating probes are silent by design, and the architecture survives with no inter-probe link at all. But a lineage whose beacon rate falls without a declared cause is losing the property that makes it more than a set of isolated nodes — and losing it for the same reasons it would lose R_eff.

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